

Topology

Home Work 3

Student: 26-712-224

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Exercise 1

Let $(X, \mathcal{U})$ be a topological space, and let $\mathcal{U}_Y$ be the induced topology of $(X, \mathcal{U})$ on $Y \subset X$. Show that:

i) If $\mathcal{B}$ is a base of $(X, \mathcal{U})$, then

$$\mathcal{B}_Y = \{V \cap Y : V \in \mathcal{B}\}$$

is a base of $(Y, \mathcal{U}_Y)$.

ii) Every closed set in $(Y, \mathcal{U}_Y)$ is closed in $(X, \mathcal{U})$ if and only if Y is closed in $(X, \mathcal{U})$.

Answer:

(i) Let $\mathcal{B}_x$ be a basis of X and let $Y \subset X$ and define

$$\mathcal{B}_y = \{Y \cap B \mid B \in \mathcal{B}_x\}$$

To show that $\mathcal{B}_y$ is a basis for Y we need to demonstrate two things:

(a) $\forall y \in Y, \exists U \in \mathcal{B}_y$ s.t. $y \in U$

Well, take $y \in Y$. As $Y \subset X$ and $\mathcal{B}_x$ is a basis for X , $\exists B_y \in \mathcal{B}_x$ s.t. $y \in B_y$.
Clearly

$$U = B_y \cap Y \in \mathcal{B}_y$$

and $y \in U$.

(b) Let $U_1, U_2 \in \mathcal{B}_y$, we need to demonstrate that:
given $y \in U_1 \cap U_2$, $\exists U_3 \in \mathcal{B}_y$ s.t.

$$y \in U_3 \subset U_1 \cap U_2.$$

Take $U_1 = Y \cap B_1$, $U_2 = Y \cap B_2$ where $B_1, B_2 \in \mathcal{B}_x$. Then:

$$\begin{aligned} U_1 \cap U_2 &= (Y \cap B_1) \cap (Y \cap B_2) \\ &= Y \cap (B_1 \cap B_2). \end{aligned}$$

As $\mathcal{B}_x$ is a basis for X and $y \in X$ $\exists B_3 \in \mathcal{B}_x$ s.t. $y \in B_3 \subset B_1 \cap B_2$.

$$\begin{aligned} \Rightarrow y \in Y \cap B_3 &= U_3 \subset Y \cap (B_1 \cap B_2) \\ &= (Y \cap B_1) \cap (Y \cap B_2) \\ &= U_1 \cap U_2. \end{aligned}$$

Points (a) and (b) demonstrate that the $\mathcal{B}_y$ is a basis for Y .

(ii)

Exercise 2

i) Let $\mathcal{U}_1$ and $\mathcal{U}_2$ be two topologies on the same space X , and let $\mathcal{B}_i \subset \mathcal{P}(X)$ be a base of $\mathcal{U}_i$, $i = 1, 2$. Show that $\mathcal{U}_1 = \mathcal{U}_2$ if and only if the following two properties are satisfied:

- a) for every $U_1 \in \mathcal{B}_1$ and every $x \in U_1$, there exists $U_2 \in \mathcal{B}_2$ such that $x \in U_2 \subset U_1$;
- b) for every $U_2 \in \mathcal{B}_2$ and every $x \in U_2$, there exists $U_1 \in \mathcal{B}_1$ such that $x \in U_1 \subset U_2$.

ii) Let $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be points in $\mathbb{R}^2$. Consider the distance

$$d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|.$$

Prove that the topology on $\mathbb{R}^2$ induced by the distance d_1 is the same as the Euclidean topology on $\mathbb{R}^2$.

Hint: Recall from Problem Sheet 1: what is the shape of an open ball for d_1 ?

Answer:

(i) Given X with topologies $\mathcal{U}_1$ & $\mathcal{U}_2$ which have bases $\mathcal{B}_1$ & $\mathcal{B}_2$ respectively. So that:

$$\begin{aligned}\mathcal{U}_1 &= \langle \mathcal{B}_1 \rangle \\ \mathcal{U}_2 &= \langle \mathcal{B}_2 \rangle.\end{aligned}$$

Take any $U \in \mathcal{B}_1$. Then for each $x \in U$ there is a $V_x \in \mathcal{B}_2$ s.t. $x \in V_x \subset U$ - from part (a).

Hence:

$$U = \bigcup_{x \in U} V_x \in \mathcal{U}_2.$$

Let $T \in \mathcal{U}$. Then, $\exists$ index I s.t.

$$\begin{aligned}T &= \bigcup_{i \in I} U_i \quad \text{each } U_i \in \mathcal{B}_1. \\ &= \bigcup_{i \in I} \bigcup_{x \in U_i} V_x \in \mathcal{U}_2.\end{aligned}$$

Since the 'double union' is still a union of open sets in $\mathcal{U}_2$. Hence

$$T \in \mathcal{U}_1 \Rightarrow T \in \mathcal{U}_2 \Rightarrow \mathcal{U}_1 \subset \mathcal{U}_2.$$

Part (b) allows the reverse argument to be made - i.e. $\mathcal{U}_2 \subset \mathcal{U}_1$.

Both arguments demonstrate that $\mathcal{U}_1 = \mathcal{U}_2$.

□

(ii) Assume now that $\mathcal{U}_1 = \mathcal{U}_2$ (i.e. $\langle \mathcal{B}_1 \rangle = \langle \mathcal{B}_2 \rangle$). Take any $U \in \mathcal{B}_1$. Then $U \in \mathcal{U}_1 \Rightarrow U \in \mathcal{U}_2$. So, there exists an index set I s.t.

$$U = \bigcup_{i \in I} V_i \quad (\text{each } V_j \in \mathcal{B}_2). \quad (*)$$

Eqn $(*) \Rightarrow$ given $x \in U$, $\exists j \in I$ s.t. $x \in V_j$. Also from $(*)$:

$$x \in V_j \subset U.$$

This covers condition (a). Reversing the roles of $\mathcal{U}_1, \mathcal{U}_2$ gives (b)

□

Exercise 3

Which of the following topological spaces are connected?

- i) $\{(x, y) \in \mathbb{R}^2 : |y| = x^2\}$, with the topology induced by the Euclidean distance on $\mathbb{R}^2$;
- ii) $\{(x, y) \in \mathbb{R}^2 : y = 1/|x|\}$, with the topology induced by the Euclidean distance on $\mathbb{R}^2$;
- iii) the annulus

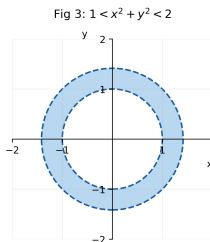
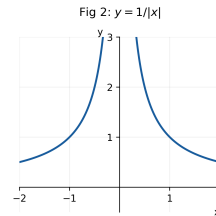
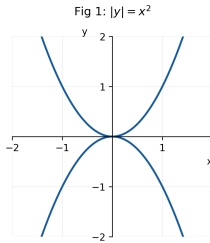
$$A(b, c) = \{(x, y) \in \mathbb{R}^2 : b < x^2 + y^2 < c\},$$

where $0 < b < c$, with the topology induced by the Euclidean distance on $\mathbb{R}^2$;

- iv) $\mathbb{R} \setminus \mathbb{Q}$, with the topology induced by the Euclidean distance on $\mathbb{R}$.

Answer:

The graphs for the 1st 3 spaces is shown below - on the annulus I have used $a = 1, b = 2$.



I hope it is OK to use mainly descriptive solutions for this problem.

(i) This set is path connected - hence connected!

(ii) I assume that we are excluding the point where $x = 0$. Here we can split $\mathbb{R}^2$ into two (open) half-planes (either side of $x = 0$). They don't overlap and they contain the complete set.

(iii) The annulus when intersected with open sets of $\mathbb{R}^2$ give connected sets of the annulus.

Hence, the annulus is also connected.

(iv) Take $S = \mathbb{R} \setminus \mathbb{Q}$. Then consider the two sets in S defined by:
 $U_1 = (-\infty, 1) \cap S$ and $U_2 = (1, \infty) \cap S$ - part of the induced topology.

It is clear that: U_1, U_2 are both not empty, $U_1 \cap U_2 = \emptyset$ and that $S = U_1 \cup U_2$.
We therefore conclude that $S = \mathbb{R} \setminus \mathbb{Q}$ is NOT connected.

Exercise 4

- a) Let $(X, \mathcal{U})$ be a connected topological space. Show that the only clopen (both closed and open) sets in X are $\emptyset$ and X .
- b) Let X be any set. Find the connected subsets of $(X, \mathcal{U})$ in each of the following cases:
- $\mathcal{U} = \{\emptyset, X\}$ is the trivial topology;
 - $\mathcal{U} = \mathcal{P}(X)$ is the discrete topology.

Note: A subset Y of $(X, \mathcal{U})$ is called connected if $(Y, \mathcal{U}_Y)$, with the induced topology, is a connected topological space.

Answer:

(a) $(X, \mathcal{U})$ connected means $\nexists U, V \subset X$ (both open) s.t.

- $U \neq \emptyset$ and $V \neq \emptyset$
- $U \cap V = \emptyset$
- $U \cup V = X$.

Assume $\exists S \subset X$ which is "clopen" and $\emptyset \neq S \neq X$. Then

- S is open
- $X \setminus S$ is open (as S is "clopen").
- $S \cap (X \setminus S) = \emptyset$ - must hold by definition of complement.
- $S \cup (X \setminus S) = X$ - must hold by definition of complement.

Contradiction. Hence, no such S exists.

(b) I will treat each topology in turn:

(i)

$$\begin{aligned} \mathcal{U} &= \{\emptyset, X\} \\ \Rightarrow \mathcal{U}_Y &= \{\emptyset, Y\}. \end{aligned}$$

It is not possible to find two open sets in $\mathcal{U}_Y$ s.t. $U \neq \emptyset$, $V \neq \emptyset$ with $U \cap V = \emptyset$ and $Y = U \cup V$.

Hence: Y is connected.

(ii)

$$\begin{aligned}\mathcal{U} &= \mathcal{P}(X) \\ \Rightarrow \mathcal{U}_Y &= \mathcal{P}(Y)\end{aligned}$$

The discrete topology is such that all sets are both open & closed (i.e. clopen).
Using the result from part (i), we must conclude that:

Y is disconnected.

This only applies if Y contains more than 1 point!!

Exercise 5

Let $(X, \mathcal{U})$ be a connected topological space. Prove that:

- i) if $f : X \rightarrow \mathbb{Z}$ is continuous, where $\mathbb{Z}$ carries the discrete topology, then f is constant;
- ii) if $f : X \rightarrow \mathbb{Q}$ is continuous, where $\mathbb{Q}$ carries the topology induced by the Euclidean distance on $\mathbb{R}$, then f is constant.

Now consider a continuous function $h : [0, 1] \rightarrow [0, 1]$.

- iii) Use part (i) to prove that h has a fixed point.

Hint: For part (iii), you may want to consider the function

$$x \mapsto \frac{h(x) - x}{|h(x) - x|}.$$

Where is it well-defined? Is it continuous?

Answer:

(i) $(X, \mathcal{U})$ is a topological space which is connected and $f : X \rightarrow \mathbb{Z}$ continuous with $\mathbb{Z}$ having the discrete topology.

Then, for every $z \in \mathbb{Z}$, we have that $\{z\}$ is an open set (with the discrete topology) and as f is continuous: $f^{-1}(\{z\})$ is open in X . Consequently, for any $T \subset \mathbb{Z}$:

$$f^{-1}(T) \in \mathcal{U}$$

Now let f take at least two distinct values on $\mathbb{Z}$. Let one such value be k .

Then, let $U = \{k\}$ - clearly U is open in $\mathbb{Z}$.

Also let $V = \{z \in \mathbb{Z} \mid z \neq k\}$ - also clearly open in $\mathbb{Z}$.

Then:

$$\begin{aligned} f^{-1}(U) = \bar{U} & \text{ is open in } X. \\ f^{-1}(V) = \bar{V} & \text{ is open in } X. \end{aligned}$$

Obviously:

$$\begin{aligned} \bar{U} \neq \emptyset \quad \& \quad \bar{V} \neq \emptyset. \\ \bar{U} \cap \bar{V} &= \emptyset \\ \text{and } U \cup V &= X. \end{aligned}$$

Contradiction, as this means that X is not connected. Hence, it is not possible for f to be non-constant.

□

(ii) Assume $f : X \rightarrow \mathbb{Q}$ is not constant. Then $\exists x_1, x_2 \in X$ s.t. $f(x_1) = q_1, f(x_2) = q_2$ where $q_1, q_2 \in \mathbb{Q}$ and $q_1 \neq q_2$.

Assume $q_1 < q_2$. Take any irrational number between q_1 & q_2 – e.g.

$$i = q_1 + \frac{1}{\sqrt{2}}(q_2 - q_1)$$

Then $O_1 = (-\infty, i)$ & $O_2 = (i, \infty)$ are both open in $\mathbb{R}$

$$\Rightarrow O_1 \cap \mathbb{Q} = R_1 \quad \& \quad O_2 \cap \mathbb{Q} = R_2$$

are both open in $\mathbb{Q}$ (with its induced topology).

As f is continuous $f^{-1}(R_1)$ & $f^{-1}(R_2)$ are both open in X , both are non-empty. Now, as $R_1 \cup R_2 = \mathbb{Q}$

$$\begin{aligned} f^{-1}(\mathbb{Q}) &= X \\ &= f^{-1}(R_1 \cup R_2) \\ &= f^{-1}(R_1) \cup f^{-1}(R_2) \end{aligned} \tag{5.1}$$

Also: $R_1 \cap R_2 = \emptyset$ and hence:

$$\begin{aligned} f^{-1}(R_1 \cap R_2) &= f^{-1}(R_1) \cap f^{-1}(R_2) \\ &= f^{-1}(\emptyset) \\ &= \emptyset. \end{aligned}$$

The two open and non-empty sets $f^{-1}(R_1)$ & $f^{-1}(R_2)$ suffice to show that X is **not** connected.

Contradiction! Hence, f MUST be constant.

(iii) We are given a continuous function $h : [0, 1] \rightarrow [0, 1]$ we define a new function:

$$f(x) = \frac{h(x) - x}{|h(x) - x|}$$

If we assume that h has no fixed pt in $[0, 1]$ then f is always well defined and continuous on $[0, 1]$ taking values of ± 1 .

Hence

$$f : [0, 1] \rightarrow \{-1, +1\} \quad (\subset \mathbb{Z}).$$

$\{-1, +1\}$ has the discrete topology (obtained when we intersect this set with the Euclidean topology defined on $\mathbb{R}$).

We know that under the Euclidean topology the set $[0, 1]$ is connected. The codomain of f is a discrete topological space which means that the function MUST be constant if it is continuous.

But this means that:

$$\frac{h(x) - x}{|h(x) - x|} \text{ is constant } \forall x \in [0, 1].$$

But this clearly isn't true; since $f(0) = 1; f(1) = -1$ must hold. So, our assumption of no fixed points for h is false. I.e.

$$\exists x \in [0, 1] \quad \text{s.t.} \quad h(x) = x.$$

□

Exercise 6

Let $(X, \mathcal{U})$ be a topological space. The space X is called *locally connected at $x \in X$* if every neighborhood of x contains an open set V with $x \in V$ such that V , equipped with the topology induced by X , is connected. The space X is *locally connected* if it is locally connected at every point. Replacing “connected” by “path-connected” gives the corresponding notions of *locally path-connected at x* and *locally path-connected*.

- a) **The topologist’s sine curve.** Let $S \subset \mathbb{R}^2$ have the induced topology and be defined by

$$S = (\{0\} \times [-1, 1]) \cup \{(x, \sin(1/x)) : x \in (0, 1]\}.$$

Show that S is connected but not locally connected.

- b) **The topologist’s comb.** Let $C \subset \mathbb{R}^2$ have the induced topology and be defined by

$$C = (\{0\} \times [0, 1]) \cup ([0, 1] \times \{0\}) \cup \bigcup_{n \in \mathbb{N}} (\{1/n\} \times [0, 1]).$$

Show that C is path-connected but not locally path-connected.